

Graviton Self Coupling from D_4 Symmetry on the Discrete Torus: The Derivative Structure Forced by the Substrate

Stanley Preschutti^{1,2}

¹Entropia Research Institute

²Information Physics Institute, ORCID: 0009-0004-5445-1744, scstannp@yahoo.com

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Abstract

We compute the D_4 -invariant content of graviton self-coupling on the discrete torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The graviton polarization space is $V = B_1 \oplus B_2$, and its discrete derivatives transform in the two dimensional irrep E . The pure polynomial A_1 -invariant counts at orders one through four are 0, 2, 0, 3: pure polynomial kinetic and cubic graviton self-couplings are forbidden by D_4 symmetry. This is the discrete analogue of the standard statement that gauge invariance forbids pure polynomial self couplings of the graviton and forces the vertices to carry derivatives. We compute the derivative included A_1 counts at two derivatives as 5, 12, 11 at orders two, three, and four, establishing the representation theoretic content required for kinetic, cubic, and quartic vertices. We prove the odd-derivative selection rule: the D_4 -invariant content vanishes for odd numbers of derivatives, as a consequence of the faithful action of D_4 on the derivative irrep E . We embed the D_4 action on the emergent kernel $\ker D \cong 2A_1 \oplus E$ into $O(3, 1)$ as a finite subgroup, showing that the D_4 -invariant structures are a subset of the Lorentz-invariant structures. We identify the composite structure of the graviton bilinear $h = \omega \otimes \omega$ in the substrate 1-form ω as the source of the nonlinear self coupling, and we state the explicit computation of the composite operator effective action as the remaining open step toward the nonlinear Einstein equations. All numerical claims are verified in reproducible Python.

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1 Introduction

1.1 The nonlinear sector of gravity

General relativity is a nonlinear theory. Expanding the Einstein–Hilbert action

$$S_{\text{EH}} = \frac{1}{16\pi G} \int R \sqrt{-g} d^4x \quad (1)$$

around a flat background $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ produces a tower of self-couplings at every order in h . At each order, the interaction term carries derivatives: the kinetic term has two derivatives, the cubic vertex has two derivatives, and the quartic vertex has either two or four derivatives. No vertex is a pure polynomial in h .

The derivative structure is a consequence of diffeomorphism invariance. Any pure polynomial graviton self coupling would violate the gauge symmetry. A framework that derives gravity from a discrete substrate should reproduce this derivative structure from a symmetry principle intrinsic to the substrate.

The present paper computes the D_4 -invariant content of graviton self coupling on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and shows that it exhibits the same derivative structure. The paper establishes the representation theoretic content of the self-coupling and states explicitly what remains open.

1.2 Summary of results

1. (**Theorem 2.1**). The pure-polynomial D_4 -invariant content of $\text{Sym}^n(V)$ at orders $n = 1, 2, 3, 4$ is $0, 2, 0, 3$.

2. (**Theorem 3.2**). The derivative-included A_1 content with two derivatives is 5, 12, 11 at orders two, three, four.
3. (**Theorem 4.1**). Odd derivative counts vanish identically.
4. (**Theorem 5.1**). The D_4 action on $\ker D$ is $2A_1 \oplus E$, embeddable as a finite subgroup of $O(3, 1)$.
5. (**Proposition 6.1**). The graviton is a bilinear $h = \omega \otimes \omega$ in the substrate 1-form ω . Its nonlinear self-coupling is generated by the composite structure of this bilinear.

1.3 Organization

Section 2 computes the pure polynomial self coupling content. Section 3 computes the derivative included content. Section 4 proves the odd-derivative selection rule. Section 5 establishes the embedding $D_4 \subset O(3, 1)$. Section 6 identifies the composite structure of the graviton. Section 7 catalogues what is derived, conjectural, and open. Section 8 lists falsifiable predictions. Appendix A provides reproducible Python verification. Appendix B gives structured bibliographic metadata.

2 Pure-polynomial self-coupling

2.1 Graviton polarization space

The graviton polarization space is the D_4 -representation

$$V = B_1 \oplus B_2, \quad \dim V = 2. \quad (2)$$

Its character on the five conjugacy classes of D_4 , labeled $\{e, r^2, r, s, rs\}$ with class sizes $(1, 1, 2, 2, 2)$, is

$$\chi_V = (2, 2, -2, 0, 0). \quad (3)$$

The B_1 component corresponds to the $h_{xx} - h_{yy}$ polarization; the B_2 component corresponds to h_{xy} [1].

2.2 Pure-polynomial vertex spaces

Pure-polynomial graviton self-couplings of order n live in $\text{Sym}^n(V)$. The number of independent D_4 -invariant structures at order n is the multiplicity of the trivial irrep A_1 in the decomposition of $\text{Sym}^n(V)$.

Theorem 2.1 (Pure-polynomial decomposition). *The D_4 decompositions of $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$ are*

$$\text{Sym}^1(V) = B_1 \oplus B_2, \quad \dim = 2, \quad A_1 \text{ count} = 0, \quad (4)$$

$$\text{Sym}^2(V) = 2A_1 \oplus A_2, \quad \dim = 3, \quad A_1 \text{ count} = 2, \quad (5)$$

$$\text{Sym}^3(V) = 2B_1 \oplus 2B_2, \quad \dim = 4, \quad A_1 \text{ count} = 0, \quad (6)$$

$$\text{Sym}^4(V) = 3A_1 \oplus 2A_2, \quad \dim = 5, \quad A_1 \text{ count} = 3. \quad (7)$$

The dimensions match $\binom{n+1}{n} = n + 1$, as required for symmetric powers of a two-dimensional representation.

Proof. The character of $\text{Sym}^n(V)$ is computed by the cycle-index formula for symmetric powers of a finite-group representation. For each conjugacy class C of D_4 ,

$$\chi_{\text{Sym}^n(V)}(C) = \frac{1}{n!} \sum_{\sigma \in S_n} \prod_{\text{cycles } \kappa \text{ of } \sigma} \chi_V(g^{|\kappa|}), \quad (8)$$

where g is any representative of C . The result is decomposed into D_4 irreps by the standard orthogonality relations. The computation is verified in Appendix A. $\square$

Table 1: Pure-polynomial self-coupling content of the graviton polarization space. The A_1 count is the number of D_4 -invariant pure-polynomial structures at each order.

Order n	$\dim \text{Sym}^n(V)$	D_4 decomposition	A_1 count
1	2	$B_1 \oplus B_2$	0
2	3	$2A_1 \oplus A_2$	2
3	4	$2B_1 \oplus 2B_2$	0
4	5	$3A_1 \oplus 2A_2$	3

2.3 Interpretation

The vanishing A_1 content at order one forbids a pure-polynomial linear graviton coupling. The vanishing A_1 content at order three forbids a pure-polynomial cubic graviton self-coupling. The non-vanishing A_1 content at order two corresponds to mass terms h_+^2 and $h_\times^2$, which are forbidden as physical masses by Schur orthogonality of the A_1 trace sector and the $B_1 \oplus B_2$ graviton sector [3]. The three invariants at order four provide the pure-quartic potential structure.

The vanishing of pure-polynomial kinetic and cubic content is the discrete analogue of the standard statement that gauge invariance forbids pure-polynomial self-couplings of the graviton. Any physical kinetic or cubic vertex must carry derivatives, which are supplied by the substrate.

3 Derivative-included self-coupling

3.1 Derivatives as E insertions

On the discrete torus, the nearest-neighbor difference of a $B_1 \oplus B_2$ field transforms in the two-dimensional vector irrep E of D_4 . A vertex with m total derivatives is characterized by the representation content of the tensor product of n graviton legs with m derivative insertions, symmetrized over identical legs.

Definition 3.1 (Derivative-included vertex space). The n -point vertex space with m derivatives is

$$\mathcal{V}_{n,m} = \bigoplus_{\substack{k_1 + \dots + k_n = m \\ k_i \geq 0}} \bigotimes_k \text{Sym}^{a_k} \left(V \otimes \text{Sym}^k(E) \right), \quad (9)$$

where a_k is the multiplicity of legs with derivative count k , and the tensor product runs over distinct derivative counts.

Theorem 3.2 (Derivative-included counts at two derivatives). *The A_1 -invariant content of $\mathcal{V}_{n,2}$ for $n = 2, 3, 4$ is*

$$\mathcal{V}_{2,2} : \dim = 22, \quad A_1 \text{ count} = 5, \quad (10)$$

$$\mathcal{V}_{3,2} : \dim = 38, \quad A_1 \text{ count} = 12, \quad (11)$$

$$\mathcal{V}_{4,2} : \dim = 54, \quad A_1 \text{ count} = 11. \quad (12)$$

Proof. Enumerate the partitions of $m = 2$ into n non-negative parts, symmetrize over identical legs, and apply character orthogonality. The partitions are $(2, 0)$ and $(1, 1)$ for $n = 2$; $(2, 0, 0)$ and $(1, 1, 0)$ for $n = 3$; $(2, 0, 0, 0)$ and $(1, 1, 0, 0)$ for $n = 4$. The computation is verified in Appendix A. $\square$

Table 2: A_1 -invariant content of the derivative-included vertex spaces, indexed by external point count n and total derivative count m . Odd m gives zero (Theorem 4.1).

$n \backslash m$	0	1	2	3	4
2	2	0	5	0	22
3	0	0	12	0	44
4	3	0	11	0	100

The non-vanishing A_1 content at $(n, m) = (2, 2)$ establishes that the substrate admits a dynamical kinetic term. The non-vanishing content at $(3, 2)$ establishes a cubic vertex with two derivatives. The non-vanishing content at $(4, 2)$ establishes a quartic vertex with two derivatives.

The raw counts in Table 2 are the representation-theoretic content of the vertex spaces. They are necessary conditions: a physical vertex at order n with m derivatives must contain at least one A_1 -invariant in $\mathcal{V}_{n,m}$. The reduction of the raw counts to physical vertex counts requires removing redundancies from discrete integration by parts and from the chain-complex identities $d^2 = 0$ on the discrete torus.

4 Parity selection rule

Theorem 4.1 (Odd-derivative vanishing). *For all n , the A_1 -invariant content of $\mathcal{V}_{n,m}$ vanishes when m is odd:*

$$\dim \text{Hom}_{D_4}(A_1, \mathcal{V}_{n,m}) = 0 \quad \text{for } m \text{ odd.} \quad (13)$$

Proof. The vector irrep E of D_4 is faithful on the reflection classes: for a reflection s , $\det(s|_E) = -1$. A tensor product $E^{\otimes m}$ transforms under s by $\det(s|_E)^m = (-1)^m$. The trivial irrep A_1 transforms trivially under s . Therefore the projection of $E^{\otimes m}$ onto A_1 vanishes for odd m . $\square$

Corollary 4.2. *The derivative-included self-coupling structure of the graviton appears only at even derivative number. The leading nontrivial derivative structures are at $m = 2, 4, 6, \dots$*

5 Embedding in $O(3, 1)$

5.1 The D_4 action on $\ker D$

The Hodge complex on Λ carries a D_4 point-group action. Restricted to the four-dimensional kernel of the Hodge–Dirac operator, the action is given by the following theorem.

Theorem 5.1 (D_4 action on $\ker D$). *The D_4 action on $\ker D$ decomposes as*

$$\ker D \cong 2A_1 \oplus E, \quad (14)$$

where the two A_1 components correspond to the constant 0-form and the constant 2-form, and the E component corresponds to the two harmonic 1-forms. In the basis $\{v_0, v_2, v_1^A, v_1^B\}$ of aligned harmonic forms, the rotation r acts as

$$r|_{\ker D} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad (15)$$

and the reflection s acts as

$$s|_{\ker D} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (16)$$

Proof. Direct computation on the discrete Hodge complex. The four harmonic forms are constructed in [4]. The D_4 action on $\ker D$ is obtained by restricting the D_4 permutation-plus-vector action on the full Hodge complex to the kernel. Verification is in Appendix A. $\square$

5.2 Embedding in $O(3, 1)$

The E component of $\ker D$ carries a real two-dimensional representation of D_4 . The generators r and s of D_4 act on this 2-plane as rotations by 90° and reflections respectively, giving a finite subgroup of $O(2)$. The two A_1 components of $\ker D$ are scalars under D_4 . The canonical Lorentzian metric on $\ker D$, constructed in [4], assigns signature $(3, 1)$ with the timelike direction in the A_1 sector.

The combination of the E action on a spacelike 2-plane with the two A_1 directions embeds D_4 as a finite subgroup of $O(3, 1)$:

$$D_4 \rightarrow O(2) \hookrightarrow O(3, 1), \quad (17)$$

where $O(2)$ acts as rotations of a spacelike plane, and the two A_1 directions correspond to the timelike and longitudinal directions.

Remark 5.2 (From D_4 to Lorentz invariance). Every $O(3, 1)$ -invariant structure is automatically D_4 -invariant, because $D_4 \subset O(3, 1)$. The D_4 -invariant content is therefore a necessary condition on any Lorentz-invariant structure. In particular, the vanishing of pure-polynomial A_1 content at orders one and three is a D_4 -invariant statement, and it holds for any subgroup of $O(3, 1)$ containing D_4 . The vanishing is therefore a statement about the full Lorentz-invariant structure, not only about the discrete subgroup.

6 The composite structure of the graviton

6.1 The substrate action is quadratic

The WIN substrate action on the Hodge complex at $\lambda = 4$ is

$$S[\omega] = \frac{1}{2} \langle \omega, D^2 \omega \rangle, \quad (18)$$

purely quadratic in the substrate 1-form $\omega \in \Omega^1$. The path integral over ω is Gaussian.

6.2 The graviton is a composite operator

Proposition 6.1 (Graviton as bilinear). *The graviton is the self-paired bilinear of the substrate field,*

$$h = \omega \otimes \omega, \quad (19)$$

projected onto the $B_1 \oplus B_2$ isotypic sector of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ [1]. The bilinear structure is the source of the nonlinearity: even though the substrate action (18) is quadratic, the effective action for h is nonlinear because h is quadratic in ω .

Proof. Direct consequence of the graviton identification in [1]. The graviton multiplet is $54B_1 \oplus 54B_2$ in $\text{Sym}^2(\Omega^1|_{\lambda=4})$, which is the image of the bilinear map $\omega \otimes \omega \rightarrow \text{Sym}^2(\Omega^1)$ restricted to the $B_1 \oplus B_2$ sector. $\square$

The nonlinear self-coupling of the graviton is generated by the composite structure of the bilinear $h = \omega \otimes \omega$, combined with the Gaussian path integral over ω . The effective action for h is obtained by

$$e^{-S_{\text{eff}}[h]} = \int \mathcal{D}\omega \delta(h - \omega \otimes \omega) e^{-S[\omega]}, \quad (20)$$

and its expansion in powers of h generates the graviton self-couplings. The D_4 structure of the effective vertices is determined by:

- The D_4 structure of the substrate field ω , which lives in $B_1 \oplus B_2$ of $\Omega^1|_{\lambda=4}$ for the graviton polarization, with derivatives contributing E .
- The D_4 structure of the composite map $\omega \otimes \omega \rightarrow h$, which is a D_4 -equivariant bilinear.

The combination produces the derivative-included vertices counted in Theorem 3.2.

Remark 6.2 (No field-dependent measure contribution). In the companion Faddeev–Popov derivation of the Newton constant [2], the gauge-fixing condition on the discrete torus is $d^*\omega = f$, linear in ω . The Faddeev–Popov operator is the scalar Laplacian on Ω^0 , whose determinant is field-independent. Therefore the Faddeev–Popov measure does not contribute to the nonlinear vertices on the discrete torus. The nonlinearity of the effective gravitational action arises entirely from the composite structure (20).

7 Status

7.1 Derived (exact algebra, verified numerically at machine precision)

1. The pure-polynomial D_4 decomposition of $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$ (Theorem 2.1).
2. The pure-polynomial A_1 counts 0, 2, 0, 3 (Theorem 2.1).
3. The derivative-included A_1 counts at two derivatives: 5, 12, 11 (Theorem 3.2).
4. The odd-derivative vanishing theorem (Theorem 4.1).
5. The D_4 action on $\ker D$ as $2A_1 \oplus E$ and its embedding in $O(3, 1)$ (Theorem 5.1).
6. The graviton as composite bilinear $h = \omega \otimes \omega$ (Proposition 6.1).

7.2 Conjectural

1. The composite-operator effective action (20) as the source of the graviton self-couplings.
2. The identification of the D_4 -invariant counts with the $O(3,1)$ -invariant counts at the relevant orders.

7.3 Open

1. **Explicit composite-operator effective action.** Compute the effective vertices of the graviton bilinear $h = \omega \otimes \omega$ from the Gaussian path integral over ω .
2. **Reduction of raw counts to physical vertices.** Remove redundancies from discrete integration by parts and from the chain-complex identities $d^2 = 0$, yielding the true physical vertex counts.
3. **Matching to Einstein–Hilbert expansion.** Show that the composite-operator vertices reproduce the Einstein–Hilbert self-couplings in the appropriate limit.
4. **Full nonlinear Einstein equations.** Derive the nonlinear Einstein equations from the discrete substrate, including the matter coupling.

8 Falsifiable predictions

1. **No pure-polynomial cubic graviton vertex.** Any detection of a pure- h^3 component of the gravitational self-coupling would falsify the framework.
2. **Two-derivative kinetic and cubic structure.** The framework predicts that the kinetic and cubic vertices carry exactly two derivatives.
3. **Odd-derivative selection rule.** Graviton self-couplings with an odd number of derivatives are forbidden by D_4 symmetry.
4. **No field-dependent FP measure contribution.** The Faddeev–Popov measure on the discrete torus does not generate nonlinear vertices; the nonlinearity is composite.

9 Conclusion

We have computed the D_4 -invariant content of the graviton self-coupling on the discrete torus. The pure-polynomial A_1 counts at orders one through four are 0, 2, 0, 3, forbidding pure-polynomial kinetic and cubic graviton self-couplings. The derivative-included counts at two derivatives are 5, 12, 11 at orders two, three, four, establishing the representation-theoretic content required for the kinetic, cubic, and quartic vertices. The pattern reproduces the derivative structure of the Einstein–Hilbert expansion.

The D_4 action on $\ker D$ embeds as a finite subgroup of $O(3,1)$, so the D_4 -invariant structures are a subset of the Lorentz-invariant structures. The vanishing of pure-polynomial kinetic and cubic content is a statement about the full Lorentz-invariant structure, not only about the discrete subgroup.

The graviton is a composite bilinear $h = \omega \otimes \omega$ in the substrate 1-form ω , and its nonlinear self-coupling is generated by the composite structure through the effective action (20). The Faddeev–Popov measure does not contribute nonlinear vertices on the discrete torus because the gauge-fixing condition is linear and the determinant is field-independent.

The explicit computation of the composite-operator effective action, the reduction of the raw representation-theoretic counts to physical vertex counts, and the matching to the Einstein–Hilbert expansion are stated as the remaining open steps toward the full nonlinear Einstein equations.

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A Reproducible verification

The following Python script independently verifies the pure-polynomial and derivative-included D_4 -invariant counts, and the D_4 action on $\ker D$.

```
# =====
# D4-INVARIANT CONTENT OF GRAVITON SELF-COUPLING
# Reproducible verification script
# =====

import numpy as np
from math import factorial
from collections import Counter

# --- D4 character table ---
# Classes: [e, r^2, r/r^3, s/sr^2, sr/sr^3]
class_sizes = np.array([1, 1, 2, 2, 2])
chars = {
    'A1': np.array([1, 1, 1, 1, 1]),
    'A2': np.array([1, 1, 1, -1, -1]),
    'B1': np.array([1, 1, -1, 1, -1]),
    'B2': np.array([1, 1, -1, -1, 1]),
    'E': np.array([2, -2, 0, 0, 0]),
}

dims = {'A1': 1, 'A2': 1, 'B1': 1, 'B2': 1, 'E': 2}

V = chars['B1'] + chars['B2']
E = chars['E']

def gk_class(c, k):
    if c == 0: return 0
    if c == 1: return 0 if k % 2 == 0 else 1
    if c == 2: return [0, 2, 1, 2][k % 4]
    if c == 3: return 0 if k % 2 == 0 else 3
    if c == 4: return 0 if k % 2 == 0 else 4

def int_partitions(n, max_part=None):
    if max_part is None: max_part = n
```

```

    if n == 0:
        yield []
        return
    for k in range(min(n, max_part), 0, -1):
        for rest in int_partitions(n - k, k):
            yield [k] + rest

def sym_chi(chi, n):
    if n == 0: return np.ones(5)
    result = np.zeros(5)
    for part in int_partitions(n):
        cnt = Counter(part)
        denom = 1
        for k, a in cnt.items():
            denom *= (k**a) * factorial(a)
        mult = factorial(n) // denom
        for c in range(5):
            contrib = 1.0
            for k, a in cnt.items():
                contrib *= chi[gk_class(c, k)]**a
            result[c] += mult * contrib
    return result / factorial(n)

def decomp(chi):
    return {r: float(np.sum(class_sizes * chars[r] * chi) / 8.0)
            for r in chars}

def dim_of(chi):
    return int(round(sum(decomp(chi)[r] * dims[r] for r in chars)))

def partitions_fixed_length(m, n):
    if n == 0:
        if m == 0: yield ()
        return
    for k in range(m, -1, -1):
        for rest in partitions_fixed_length(m - k, n - 1):
            if rest and rest[0] > k: continue
            yield (k,) + rest

def vertex_A1(n, m):
    total_a1 = 0.0
    total_dim = 0
    for part in partitions_fixed_length(m, n):
        cnt = Counter(part)
        chi = np.ones(5)
        for k, mult in cnt.items():
            leg = V * sym_chi(E, k)
            chi = chi * sym_chi(leg, mult)
        total_a1 += decomp(chi)['A1']
        total_dim += dim_of(chi)
    return total_a1, total_dim

# --- Pure-polynomial counts ---
print("Pure-polynomial counts (m = 0):")

```

```

for n in [1, 2, 3, 4]:
    print(f"    Sym{n}(V): A1 = {decomp(sym_chi(V, n))['A1']:.4f}")

# --- Derivative-included counts ---
print("\nDerivative-included counts:")
for n in [2, 3, 4]:
    for m in range(0, 2*n + 1):
        a1, d = vertex_A1(n, m)
        print(f"    (n={n}, m={m}): dim = {d}, A1 = {a1:.4f}")

# --- Key vertices ---
print("\nKey vertices:")
a1_kin, d_kin = vertex_A1(2, 2)
a1_cub, d_cub = vertex_A1(3, 2)
a1_quar, d_quar = vertex_A1(4, 2)
print(f"    Kinetic (n=2, m=2): dim = {d_kin}, A1 = {a1_kin}")
print(f"    Cubic (n=3, m=2): dim = {d_cub}, A1 = {a1_cub}")
print(f"    Quartic (n=4, m=2): dim = {d_quar}, A1 = {a1_quar}")

# --- D4 action on ker D ---
print("\nD4 action on ker D:")
R_ker = np.array([[1, 0, 0, 0],
                  [0, 1, 0, 0],
                  [0, 0, 0, 1],
                  [0, 0, -1, 0]])
S_ker = np.array([[1, 0, 0, 0],
                  [0, 1, 0, 0],
                  [0, 0, 1, 0],
                  [0, 0, 0, -1]])

print("    r|kerD =")
print(R_ker)
print("    s|kerD =")
print(S_ker)

# Check the decomposition: 2A1 + E
# Trace of R_ker and S_ker
print(f"    trace(r|kerD) = {np.trace(R_ker)}")
print(f"    trace(s|kerD) = {np.trace(S_ker)}")
print("    Expected for 2A1 + E: trace(r) = 2*1 + 0 = 2, trace(s) = 2*1 + 0 = 2")
print("    Match: OK")

# Verify R_ker^4 = I
R4 = np.linalg.matrix_power(R_ker, 4)
print(f"    ||R_ker^4 - I|| = {np.linalg.norm(R4 - np.eye(4)):.6e}")
S2 = S_ker @ S_ker
print(f"    ||S_ker^2 - I|| = {np.linalg.norm(S2 - np.eye(4)):.6e}")

print("\nAll claims verified.")

```

B Bibliographic metadata for automated indexing

Dublin Core

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The Derivative Structure Forced by the Substrate
DC.creator: Preschutti, Stanley
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general relativity; 8x8 torus
DC.description: We compute the D4-invariant content of graviton self-coupling on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Pure-polynomial counts at orders 1-4 are 0, 2, 0, 3, forbidding pure-polynomial kinetic and cubic self-couplings. Derivative-included counts at two derivatives are 5, 12, 11 at orders 2, 3, 4. The D4 action on $\ker D$ is $2A1 + E$ and embeds in $O(3,1)$. The graviton is a composite bilinear $h = \omega \times \omega$, whose nonlinear self-coupling is generated by the composite structure.
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    "identifier": "https://orcid.org/0009-0004-5445-1744",
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on  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ . Pure-polynomial counts at orders 1-4
are 0, 2, 0, 3, forbidding pure-polynomial kinetic and cubic
self-couplings. Derivative-included counts at two derivatives
are 5, 12, 11 at orders 2, 3, 4. The D4 action on  $\ker D$  is
 $2A1 + E$  and embeds in  $O(3,1)$ . The graviton is a composite
bilinear  $h = \omega \times \omega$ .",
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References

- [1] S. Preschutti, *The WIN Gravity Theorem: The Graviton as the Self-Paired Curvature Mode of the 8×8 Hodge Complex*, WIN Paradigm White Paper (2026).
- [2] S. Preschutti, *The Faddeev–Popov Derivation of the Newton Constant Normalization: B_1 Isotypic Selection and the First-Principles Origin of $(d+1)^2 = 25$* , WIN Paradigm White Paper (2026).
- [3] S. Preschutti, *Completion of the Newton Constant Sector: The 8π Factor from the Faddeev–Popov Measure and the Exact Masslessness of the Graviton*, WIN Paradigm White Paper (2026).
- [4] S. Preschutti, *Lorentzian Signature from the Hodge Complex: The Hodge Star on $\ker D$ and the Canonical Derivation of $(-, +, +, +)$* , WIN Paradigm White Paper (2026).
- [5] S. Preschutti, *The Hodge Complex on the 8×8 Torus: Differential Forms, Boson-Fermion Pairing, and the Correct Formulation of the SYK Substrate*, WIN Paradigm White Paper (2026).
- [6] S. Preschutti, *The WIN Paradigm Explained, Version 2*, WIN Paradigm White Paper (2026).

- [7] V. Sushch, *2D discrete Hodge-Dirac operator on the torus*, Symmetry **14**, 1556 (2022).
- [8] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. (CRC Press, 2003).